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Predictive Analytics & Forecasting

FINAL ASSIGNMENT

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PREDICTING CUSTOMER CHURN USING TELCO CUSTOMER DATA

1 INTRODUCTION

In today's competitive telecommunications industry, the retention of the customers is just as important as customer acquisition. The cost of gaining a new customer is typically far higher than retaining an existing one, yet many companies continue to struggle with customer churn (Kotler et al., 2017). The Churning phenomenon refers to the percentage of customers who discontinue or cancel their subscription within a given period. A high churn rate can seriously affect profitability, as it significantly reduces revenue and risk increase in marketing and acquisition costs, which in long run negatively affects financial performance (Idris & Khan, 2012).

Predictive analytics is a key to finding solutions to this challenge. By analyzing historical customer data, businesses can identify the key drivers of the ongoing churn and build models that predict which customers are most likely to leave. This allows companies to take targeted action, whether it be through retention offers, service improvements, loyalty reward or offering discounts or invectives before the customer churns and cancels the subscription (Amin et al., 2017).

The goal of this project was to build predictive models that help a telecommunications company identify customers at risk of churn. Using the **Telco Customer Churn dataset** from Kaggle, we performed data cleaning, exploratory analysis, feature engineering, model development, and evaluation. The results provided both predictive insights and actionable business recommendations for reducing churn.

2 PART 1: PROBLEM DEFINITION AND DATASET SELECTION

The business problem addressed in this study is:

> *"Can we predict which customers are most likely to churn, so that the telecom provider can proactively implement strategies to improve customer retention?"*

This problem is of utmost importance as the customer churn directly affects revenue of the firm. A leaving customer not only represents the lost current income but also the potential loss of future cross-sell and upsell opportunities. Hence just by preventing this churn by even a few percentage points, large telecom firms can save millions of dollars annually.

To address this problem, the **Telco Customer Churn dataset** was selected from Kaggle. The dataset contained **7,043 records** of customers, each described by **21 features**. These features included demographic details (gender, age, senior citizen status), account details (tenure, contract type, billing method), services subscribed (internet, phone, streaming services), and the target variable: churn (Yes/No).

This dataset was well-suited for the problem because of several reasons:

1. It was directly related to the telecom industry and specifically focused on churn, the targeted business problem.
2. It contained both **categorical and numerical features** which enabled the development of a wide range of predictive models.
3. The presence of a labeled churn variable made it a supervised learning problem, which aligned perfectly with predictive analytics methods.

Thus, the dataset was both relevant and sufficient for building meaningful churn prediction models.

3 PART 2: EXPLORATORY DATA ANALYSIS

The first essential step before proceeding into building models was to clean and explore the dataset, so that the quality of given data was ensured and to gain insight into the drivers of a potential churn.

3.1 DATA CLEANING AND PREPROCESSING

During initial inspection, the column **TotalCharges** was found labelled and stored as an object instead of any numerical variable. This was possibly due to blank or space entries for some customers who continued their subscription for a very short time. These values were later converted into numeric form, and missing values were added in those spaces and blanks appropriately (Han et al., 2012).

Afterwards, categorical variables like gender, contract type, and payment method were encoded using **one-hot encoding** to make them compatible with machine learning models (James et al., 2021). Class imbalance was also checked: approximately **27% of customers churned**, while **73% did not**. This imbalance was quite significant, as it may lead to bias models toward predicting non-churns more often (He & Garcia, 2009).

3.2 STATISTICAL SUMMARIES

Descriptive statistics highlighted several important patterns:

- **Tenure** for the subscriptions ranged from 0 to 72 months. Customers with shorter tenure were disproportionately represented among churners.
- **MonthlyCharges** were anywhere from $18 to $118 and were highly variable. Churn was more common among customers who were paying higher monthly charges.
- **Contract type** showed stark differences: most churned customers were those who were renewing plans on **month-to-month contracts**, whereas long- term or binding one- and two-year contracts had much lower churn rates compared to those on monthly basis.

3.3 VISUAL EXPLORATION

Visualization provided more compelling evidence of these patterns:

- A bar plot of **churn by tenure group** confirmed that customers with less tenure such as days to months, usually below one year were the most likely to churn.
- A stacked bar chart of **contract type versus churn** showed that most of the churners were on month-to-month basis plans.
- Churn rates were also found to be higher among customers using **fiber optic internet**, suggesting possible dissatisfaction with lower or badly affected service quality.

3.4 CORRELATION AND FEATURE IMPORTANCE

Correlation analysis also showed that **TotalCharges** was proportional to tenure and monthly charges, which was expected. Initial feature importance analysis from a Random Forest model also indicated that **tenure, contract type, internet service type, payment method, and charges (monthly and total)** were the detrimental factors in the prediction of churn.

These findings provided a clear path for feature engineering and model building.

4 PART 3: FEATURE ENGINEERING AND MODEL DEVELOPMENT

4.1 FEATURE ENGINEERING

To improve predictive power, several new features were designed and introduced, such as:

- **Tenure groups**: Customers were regrouped into shorter intervals (0–12, 13–24, …, 60–72 months) to capture non-linear relationships.
- **Average charges**: A new variable was created by dividing total charges paid by length of full tenure, to get the idea of the typical monthly expenditure paid by customers over their entire subscription plan.
- Categorical variables were transformed into dummy variables using one-hot encoding.
- Numerical features were normalized and used as per requirement to ensure comparability.

4.2 PREDICTIVE MODELS

Three models were selected and implemented:

1. **Logistic Regression** – A simple and interpretable model that can provide a baseline for churn prediction.
2. **Random Forest** – An ensemble model that captures non-linear relationships and their interactions between features.
3. **XGBoost** – A gradient boosting model designed for structured data, often achieving state-of-the-art performance.

4.3 HYPERPARAMETER TUNING

Each model was then tuned and prepared for optimal performance using grid search with cross-validation. For example, the number of estimators and tree depth were optimized for Random Forest, while learning rate and maximum depth were tuned for XGBoost. Logistic Regression was also tested with different regularization strengths to ensure optimal results.

4.4 DATA PIPELINE

A preprocessing pipeline was created using scikit-learn's **Pipeline** functionality. This pipeline was necessary to ensure that the categorical encoding, feature scaling, and model training were carried out frequently, consistently and reproducibly for every applied model.

5 PART 4: MODEL EVALUATION AND BUSINESS INSIGHTS

5.1 MODEL EVALUATION

The three models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Results are summarized below:

Model	Accuracy	Precision (Churn)	Recall (Churn)	F1-Score (Churn)	ROC-AUC
Logistic Regression	0.79	0.64	0.53	0.58	0.837
Random Forest	0.79	0.67	0.41	0.51	0.840
XGBoost	0.77	0.57	0.51	0.54	0.813

While all models' performance was on par with each other, **Random Forest had the greatest ROC-AUC value (0.840)**, which implied out of all it performed better in separating non-churners and churners. While its recall was weaker, it still sacrificed predictive accuracy for interpretability as well as feature importance analysis. **Random Forest was therefore selected as the ideal model for this instance.**

5.2 INTERPRETATION OF RESULTS

Feature importance from the Random Forest model revealed the top churn drivers:

- **Tenure**: Customers with shorter tenure were more likely to churn.
- **Contract type**: Month-to-month contracts had strong correlation with churn, while the customers with one- and two-year contracts were predicted to have stayed longer and showed retention.
- **Internet service (fiber optic)**: Fiber optic customers churned more frequently, which may be associated with concerns about low quality or disturbances in stable connections.
- **Payment method (electronic check)**: Customers who were paying via electronic check had more susceptibility to churn.

- **Monthly and average charges**: High cost and expenditure were found to be linked with more dissatisfaction and higher churn rates.

5.3 Business Recommendations

Based on these findings, the following recommendations are proposed for the improvement and reduced churn:

1. **Retention campaigns**: The company should focus on new customers and within their first year of service, it is advisable to offer loyalty rewards or personalized incentives for higher retention.
2. **Contract incentives**: While selling subscriptions, salespeople should encourage customers to switch from month-to-month to longer-term contracts by offering discounts or bundled services.
3. **Service quality improvements**: company should conduct a thorough investigation regarding fiber optic service complaints and address pain points leading to dissatisfaction.
4. **Payment method promotion**: there should be offers and incentives for automatic payments or credit card billing to reduce churn among customers using electronic checks.
5. **High-value customers**: Company should introduce packages to protect customers with high monthly charges by offering exclusive benefits, bundles, or VIP support.

These actions directly address the key churn drivers and provide actionable strategies for reducing attrition.

6 Part 5: Documentation and Presentation

6.1 Code Documentation

The attached Python notebook created for this particular project is well-documented with correct markdown of cells explaining every step in the analysis done methodically. Code sections have all the details and comments about data cleaning, feature engineering, model training, and testing. The visualizations provided are well-labeled and interpreted using business terms so that they can be easily understood.

6.2 REPORT SUMMARY

This report contains the full analytics process, from problem definition to business recommendations. The main key points include:

- ***Business Problem**: Identifying and defining churn prediction as a priority for telecom companies.*
- ***Dataset**: Selected Telco Customer Churn dataset because of its relevance and completeness.*
- ***Methodology**: Followed a structured process of cleaning, EDA, feature engineering, modeling, and evaluation.*
- ***Findings**: Identified several features such as tenure, contract type, charges, internet service, and payment method as top churn predictors.*
- ***Model Selection**: Random Forest was chosen as best model based on superior ROC-AUC.*
- ***Recommendations**: Devised and offered targeted strategies to reduce churn and improve retention of the customers for longer durations.*

6.3 LIMITATIONS AND FUTURE IMPROVEMENTS

While this project achieved strong results, there were several limitations, such as:

- The dataset was **imbalanced**, thereby reducing recall for churn predictions. The possibility of using techniques such as **SMOTE resampling** or cost-sensitive learning could improve this imbalance.
- The dataset lacked unstructured data, such as **customer complaints or satisfaction surveys**, which could have provided additional churn signals at an earlier stage before ending subscriptions by customers.
- Model performance could be further enhanced by exploring other advanced algorithms like **LightGBM** or **CatBoost**.

As there is always room for improvements, in future there should be focus on addressing data imbalance, integrating new data sources, and experimenting with more advanced models.

7 CONCLUSION

This project demonstrated how predictive analytics can be used to effectively address business problems of customer churn for the telecommunication sector. Using the Telco Customer Churn dataset, we inspected different aspects of customer behavior, engineered relevant features, and built predictive models. Random Forest model emerged as the best performer in this instance and recognized tenure, contract type, internet service, and payment method as key drivers for churn (Amin et al., 2017).

The business guidance derived from these conclusions can provide telco companies sound practical action strategies to retain their customers, for instance, soliciting vulnerable segments, delivering better service quality, and rewarding long-term commitments (Idris & Khan, 2012). There were limitations, but the results showed the potency of evidence-based decision-making in preventing churn and promoting customer retention (Verbeke et al., 2014).

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